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Hardware Implementation for the Enhancement of 8-bit ADC Resolution

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Abstract—This work presents a hardware implementation of a resolution enhancement technique for an 8-bit analog-to-digital converter (ADC) using analog gain scaling and reference-based digital correction. The proposed method aims to improve the effective resolution without altering the internal architecture of the ADC. The system was first validated in simulation on Proteus and subsequently implemented on hardware using a breadboard setup. This work presents the successful design, implementation, and testing of the proposed measurement circuit. The circuit was first developed on a breadboard and experimentally verified for input voltages of 0.5 V, 1 V, 2 V, 3 V, 4 V, and 4.096 V. More than 100 readings were recorded for each input level, and statistical parameters such as Mean, Relative Error (RE), Signal-to-Noise Ratio (SNR), and Effective Number of Bits (ENOB) were calculated to evaluate system performance. After successful validation, a PCB was designed, fabricated, and fully soldered. The final hardware was interfaced with an Arduino and a 16x2 LCD for real-time display.

Index Terms—ADC, DAC, Resolution Enhancement, Gain Amplifier, Reference Voltage (Vref), Rail Splitter, LabVIEW, Hardware Implementation, PCB, LCD16x02

I. METHODOLOGY

The methodology adopted for improving the resolution of an 8-bit ADC involves the following steps:

- Analog-to-Digital Conversion (ADC):**
The input DC analog voltage V_{in} is fed to an 8-bit ADC, which performs the analog-to-digital conversion and produces a digital output D_1 .
- Digital-to-Analog Conversion (DAC):**
The digital output D_1 is converted back to its analog equivalent (V_{dac}) using an 8-bit DAC.
- Error Extraction and Amplification:**
The difference between the actual input voltage and the DAC output ($V_{in} - V_{dac}$) represents the quantization error. This error is amplified using a differential amplifier to make it more significant and easier to resolve.
- Second Stage ADC Conversion:**
The amplified error is again fed to the ADC, and the result (D_2) is stored.
- Accurate Input Voltage Estimation:**
Using both D_1 and D_2 , a more accurate estimate of the input voltage (V_{in}) is computed.

Step 1: Feed V_{in} to ADC

$$D_1 \approx \frac{V_{in}}{\Delta}, \quad \Delta = \frac{V_{ref}}{2^N}$$

Equivalent DAC output: $V_{DAC1} = D_1 \times \Delta$

Step 2: Get amplified error

$$V_o = G(V_{in} - V_{DAC1})$$

Step 3: Feed to ADC

$$D_2 = \frac{V_o}{\Delta} = \frac{G(V_{in} - V_{DAC1})}{\Delta}$$

Step 4:

$$V_{in} = V_{DAC1} + \frac{\Delta D_2}{G}$$

(This term minimizes quantization error of previous case)

New quantization error:

$$\Delta' = \frac{\Delta G - \Delta D_2}{G} = \frac{(G - D_2)\Delta}{G}$$

$$\Delta' = \frac{V_{ref}}{2^{N'}} \Rightarrow \frac{V_{ref}}{2^{N'}} = \frac{(G - D_2)\Delta}{G}$$

$N' \rightarrow$ Represents new bit-size of overall ADC.

$$\frac{V_{ref}}{2^{N'}} = \frac{(G - D_2)}{G} \Delta = \frac{(G - D_2)}{G} \frac{V_{ref}}{2^N}$$

$$2^{N'} = \frac{G \cdot 2^N}{G - D_2}$$

Taking log on both sides: $N' \log 2 = N \log 2 + \log G - \log(G - D_2)$

$$\Rightarrow N' = N + \frac{\log G - \log(G - D_2)}{\log 2}$$

$G - D_2 = 1, \log(G - D_2) \Rightarrow$ term approaches 0

For ex: if $G = 256, D_2 = 255$

Modified bit size: $N' = N + \log_2 G$

Corresponding digital output: $GD_1 + D_2$

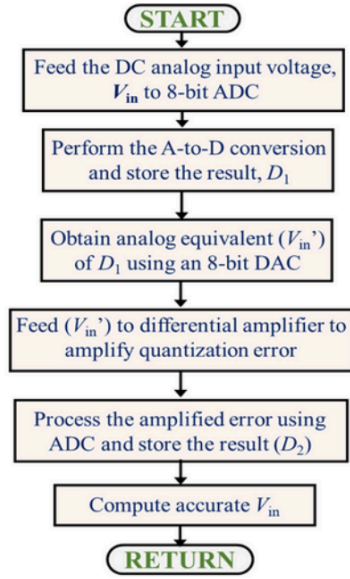


Fig. 1: Operation flowchart

II. IMPLEMENTATION AND HARDWARE MODIFICATIONS

A. The implementation is carried out using a Successive Approximation Register (SAR) ADC.

- To increase resolution, the difference $V_{in} - V_{dac}$ is amplified and fed again into the ADC.
- This approach makes quantization error measurable.
- Validation is being performed using an 8-bit SAR ADC IC 0808.

B. Key Considerations for ADC:

- The internal DAC output of ADC0808 is not available.
- To overcome this limitation, an external DAC (MCP4802) is used.
- If the internal DAC voltage is accessible, an external DAC becomes optional.

C. External split supply used to power the gain op-amp

- Rail splitter circuit was verified through simulation in Proteus, confirming proper dual supply generation.

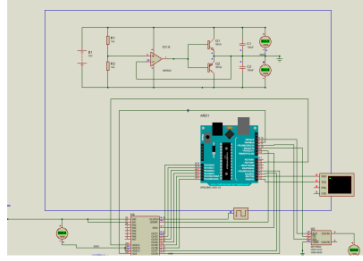


Fig. 2: Whole Proteus implementation using rail splitter

- The circuit was subsequently implemented and tested on a breadboard, and then integrated with the main system to provide $\pm$ supply to the gain amplifier.

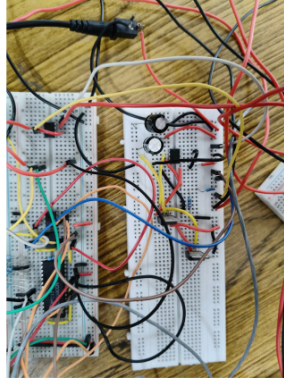


Fig. 3: Rail-Splitter circuit on breadboard

III. IMPLEMENTATION OF SCHEME ON BREADBOARD

A. Components Used

The hardware setup includes the following key components:

- Arduino Uno
- ADC0808 (8-bit SAR ADC)
- MCP4802 (8-bit dual channel SPI DAC)
- AD620 instrumentation amplifier (as a comparator with adjustable gain)
- Resistors, LEDs
- OP07 opamp (for adjusting loading errors)
- CD4053B multiplexer

B. Circuit Operation

The input voltage V_{in} is applied to channel IN0 via CD4053B multiplexer of the ADC0808. The Start of Conversion (SOC) for the ADC is triggered by a falling edge pulse on the START pin, generated by the Arduino. The End of Conversion (EOC) pin goes low during the conversion process and returns high upon completion, which is monitored by the Arduino. The Output Enable (OE) is then asserted to read the digital output.

The digital output from the first conversion is immediately fed to the MCP4802. The MCP4802 produces a voltage output which is along with the input voltage fed to the amplifier. The resulting error voltage ($V_{in} - V_{dac}$) is amplified by the AD620 instrumentation amplifier, and the gain of which is set by a single external resistor R_G and is given by:

$$G = 1 + \frac{49.4}{R_G} \quad (1)$$

The Arduino selects another channel, and the amplified error signal is routed through the CD4053B multiplexer to the ADC0808. The resulting second conversion value D_2 is read and stored.

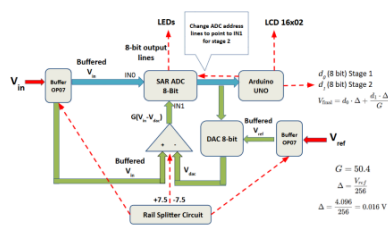


Fig. 4: Connection diagram

The MCP4802 includes an internal 2.048 V reference and a selectable output-gain stage. Setting the gain bit to GA = 0 gives $\times 1$ (full-scale = 2.048 V), and GA = 1 gives $\times 2$ (full-scale = 4.096 V). In this work we use the $\times 2$ setting, i.e. $V_{\text{FS DAC}} = 4.096 \text{ V}$.

To avoid scale-factor errors, the ADC must operate with the **same** full-scale as the DAC. Thus we set the ADC reference to an external 4.096 V. If a 5 V ADC reference is kept, a fixed scale correction is required. Also, we have used the AD620 gain such that the *maximum* expected output stays within the ADC/DAC full-scale.

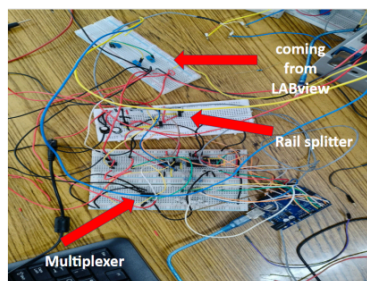


Fig. 5: Whole Breadboard implementation

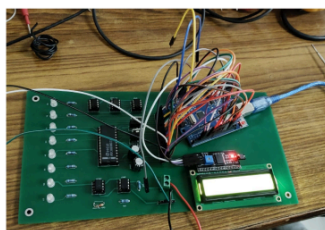


Fig. 6: Fabricated PCB

IV. SIGNAL-TO-NOISE RATIO (SNR) AND EFFECTIVE NUMBER OF BITS (ENOB)

A. Signal-to-Noise Ratio (SNR)

Signal-to-Noise Ratio (SNR) is a measure of the strength of a signal relative to background noise. It is defined as the ratio of signal power to noise power.

$$SNR = \frac{P_{\text{signal}}}{P_{\text{noise}}} \quad (2)$$

In decibels (dB), SNR is expressed as:

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \quad (3)$$

For a discrete-time signal $a(i)$ with N samples, the signal power and noise power can be written as:

$$P_{\text{signal}} = \frac{1}{N} \sum_{i=1}^N [a(i)]^2 \quad (4)$$

$$P_{\text{noise}} = \frac{1}{N} \sum_{i=1}^N [a(i) - \bar{a}]^2 \quad (5)$$

where $\bar{a}$ is the mean of the signal.

Thus, SNR in terms of discrete samples becomes:

$$SNR_{dB} = 10 \log_{10} \left(\frac{\sum_{i=1}^N [a(i)]^2}{\sum_{i=1}^N [a(i) - \bar{a}]^2} \right) \quad (6)$$

B. Quantization Noise in ADC

In an ideal N -bit Analog-to-Digital Converter (ADC), the quantization step size is given by:

$$\Delta = \frac{V_{FS}}{2^N} \quad (7)$$

where V_{FS} is the full-scale voltage.

The quantization noise is assumed to be uniformly distributed in the range $[-\Delta/2, \Delta/2]$. Therefore, the quantization noise power is:

$$P_{noise} = \frac{\Delta^2}{12} \quad (8)$$

For a full-scale sinusoidal input signal, the signal power is:

$$P_{signal} = \frac{V_{FS}^2}{8} \quad (9)$$

C. SNR of an Ideal ADC

The SNR for an ideal ADC can be derived as:

$$SNR = \frac{V_{FS}^2/8}{\Delta^2/12} \quad (10)$$

Substituting $\Delta = \frac{V_{FS}}{2^N}$:

$$SNR = \frac{3}{2} \cdot 2^{2N} \quad (11)$$

Taking logarithm in decibels:

$$SNR_{dB} = 10 \log_{10} \left(\frac{3}{2} \cdot 2^{2N} \right) \quad (12)$$

$$SNR_{dB} = 6.02N + 1.76 \quad (13)$$

D. Effective Number of Bits (ENOB)

Effective Number of Bits (ENOB) represents the actual resolution of an ADC after accounting for noise and non-idealities.

It is derived from the measured SNR as:

$$ENOB = \frac{SNR_{dB} - 1.76}{6.02} \quad (14)$$

E. Discussion

From the above equations, it is evident that:

- Each additional bit in ADC increases SNR by approximately 6.02 dB.
- The constant 1.76 dB arises due to quantization of a sinusoidal signal.
- In practical systems, ENOB is always less than the actual number of bits due to thermal noise, non-linearity, and other imperfections.

V. READINGS AND RESULTS

For each selected input voltage level (0.5 V, 1 V, 2 V, 3 V, 4 V, and 4.096 V), 100 output samples were recorded. These samples were used to evaluate the performance metrics such as mean, standard deviation, relative error (RE), signal-to-noise ratio (SNR), and effective number of bits (ENOB).

A. Experimental Relation

The final output voltage is given by:

$$V_{final} = d_0 \Delta + \frac{d_1 \Delta}{G} \quad (15)$$

B. Statistical Results

TABLE I: Statistical Performance Parameters for Different Input Voltages

Parameter	0.5 V	1 V	2 V	3 V	4 V	4.096 V
Mean	0.55889	1.05649	2.056444	3.02643	4.04647	4.09561
SD	0.000507	0.000547	0.000657	0.000660	0.000180	0.000038
RE (%)	0.624699	0.270386	0.138905	0.09410	0.01805	0.00366
SNR (dB)	60.88609	65.76161	69.96050	73.34121	75.78362	81.21324
ENOB (bits)	9.82161	10.63150	11.32899	11.89056	12.29628	13.19821

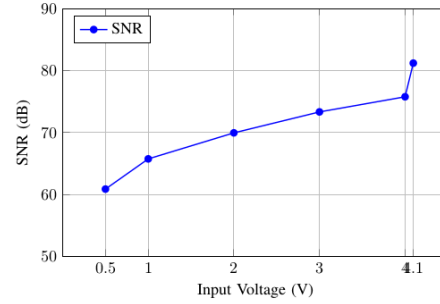


Fig. 7: SNR variation with input voltage

C. Observations

- The mean output closely matches the corresponding input voltage, indicating good accuracy of the system.
- The standard deviation is very small, showing low noise measurements.
- The Signal-to-Noise Ratio (SNR) increases with input voltage, implying improved signal quality at higher voltages.
- The Effective Number of Bits (ENOB) also increases with input voltage, indicating better resolution performance. At lower inputs, the system achieves around 9.82 bits, while near full-scale input (4.096 V), it reaches approximately 13.2 bits.

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VI. CONCLUSION

The project involves a working circuit on a hardware breadboard interfaced with an Arduino UNO. After successful validation, a PCB was designed, fabricated, and fully soldered. The final hardware was interfaced with an Arduino and a 16x2 LCD for real-time display.

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